

Model development

Oksana Chernova

30 January, 2026

Loading libraries

```
library(openxlsx)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(ggplot2)
library(ggExtra)
library(patchwork)
library(stringr)
library(knitr)
library(gtsummary)
```

Read excel sheet

```
# Set working directory
setwd("//nas.ads.mwn.de/tuma/m12/Projects/TUM MRI biopsy")

data <- read.xlsx(paste0(getwd(), "/Biopsien.Update.21.01.2026.xlsx")
                  ,sheet = "Tabelle1 (cleaned)"
                  ,detectDates = TRUE)

dim(data)

## [1] 1387  49

c(min(data$Date_Biopsy), max(data$Date_Biopsy))

## [1] "2019-11-27" "2024-09-25"
```

Data pre-processing

```
colnames(data)[c(26, 27, 28)] <- c("Experience_total", "Experience_transrectal", "Experience_perineal")

# Extract the text before the dot
extracted_text <- gsub("\\\\.*", "", colnames(data))
colnames(data) <- extracted_text

# Data pre-processing
data <- data %>%
  rename(Insurance = Insurance) %>%
  mutate(
    Insurance = factor(tolower(Insurance), levels = c("a", "p"), labels = c("Public", "Privat")),
    Age = as.numeric(Age),
    DRE = tolower(DRE),
    Biopsy_type = case_when(
      Biopsy_type %in% c("Target + Systematic", "Target+Systematic",
                        "Targeted+Systematic") ~ "Targeted+Systematic",
      TRUE ~ Biopsy_type
    ),
    Volume = ifelse(Volume %in% c("?", "Na"), NA, Volume),
    PIRADS = ifelse(PIRADS %in% c("?", "-"), NA, PIRADS),
    ISUP = ifelse(ISUP %in% c("?", "-"), NA, ISUP),
    Target_cores_taken = ifelse(Target_cores_taken %in% c("?", "-"), NA, Target_cores_taken),
    L1_region = ifelse(L1_region %in% c("?", "-"), NA, L1_region),
    L1_sector = ifelse(L1_sector %in% c("?", "-"), NA, L1_sector),
    L1_cores_taken = ifelse(L1_cores_taken %in% c("?", "-"), NA, L1_cores_taken),
    L2_PIRADS = ifelse(L2_PIRADS %in% c("?", "-"), NA, L2_PIRADS),
    L2_ISUP = ifelse(L2_ISUP %in% c("?", "-"), NA, L2_ISUP),
    # remove space from start and end of string
    Prior_neg_biopsy = str_trim(Prior_neg_biopsy),
    Biopsy_method = str_trim(Biopsy_method),
    Target_method = str_trim(Target_method),
    L1_side = str_trim(L1_side),
    L3_region = str_trim(L3_region)
  ) %>% mutate(
    Volume = as.numeric(Volume, na.strings = "NA"),
    ISUP = as.numeric(ISUP, na.strings = "NA")
  )

# PSA values with special symbols ----
# Filter for rows where the 'PSA' contains '>', '?' or ','
df_psa_sym <- data %>%
  filter(str_detect(PSA, "[>?,]") ) %>%
  select(ID, Date_Biopsy, Prior_cancer, PSA, DRE, PIRADS, ISUP, Biopsy_type, Biopsy_method)

#kable(df_psa_sym, format = 'markdown')

# Extract a value that follows a ">"
data$PSA <- ifelse(grepl("^>", data$PSA), sub("^>", "", data$PSA), data$PSA)
# Extract a value that follows a "?"
data$PSA <- gsub("\\\\?", "", data$PSA)
data$PSA <- gsub("\\\\,", ".", data$PSA)
```

```

data$PSA <- as.numeric(data$PSA, na.strings = "NA")

psa_large <- data %>%
  filter(PSA >= 100) %>%
  select(ID, Date_Biopsy, Prior_cancer, PSA, DRE, PIRADS, ISUP, Biopsy_type, Biopsy_method) %>%
  arrange(PSA)
nrow(psa_large)

## [1] 17

#kable(psa_large, format = 'markdown')

# Data processing

data <- data %>%
  mutate(PIRADScat = as.factor(case_when(
    PIRADS == 0 ~ "PI-RADS=0",
    PIRADS == 1 ~ "PI-RADS=1",
    PIRADS == 2 ~ "PI-RADS=2",
    PIRADS == 3 ~ "PI-RADS=3",
    PIRADS == 4 ~ "PI-RADS=4",
    PIRADS == 5 ~ "PI-RADS=5",
    TRUE ~ NA
  )),
  PIRADScat2 = case_when(
    PIRADS %in% 2:3 ~ "2-3",
    PIRADS == 4 ~ "4",
    PIRADS == 5 ~ "5",
    TRUE ~ NA
  ),
  PIRADScat3 = case_when(
    PIRADS %in% 2:3 ~ "2-3",
    PIRADS >= 4 ~ "4-5",
    TRUE ~ NA
  ),
  ISUPcat = case_when(
    ISUP < 2 ~ "ISUP<2",
    ISUP == 2 ~ "ISUP=2",
    ISUP > 2 ~ "ISUP>2",
    TRUE ~ NA),
  csPCa = factor(ifelse(ISUP >= 2, "csPCa", "non-csPCa"),
    levels = c("non-csPCa", "csPCa"))
)

# Potential exclusions
table(data$Prior_cancer, useNA="always")

##
##   No   Yes <NA>
## 1374   10     3

excl1 <- data %>%
  filter(Prior_cancer %in% c("Yes") | PIRADS %in% c("1", "PET CT") | is.na(PIRADScat)) %>%
  select(ID, Date_Biopsy, Prior_cancer, PSA, DRE, PIRADS, ISUP, Biopsy_type, Biopsy_method) %>%

```

```

arrange(desc(Prior_cancer), PSA)

excl2 <- data %>% filter(!(ID %in% excl1$ID) ) %>%
  filter(PSA >= 100) %>%
  select(ID, Date_Biopsy, Prior_cancer, PSA, DRE, PIRADS, ISUP, Biopsy_type, Biopsy_method)%>%
  arrange(desc(Prior_cancer), PSA)

excl <- data %>%
  filter(Prior_cancer %in% c("Yes") | PIRADS %in% c("1", "PET CT") | PSA >= 100 |
         is.na(Biopsy_type) | is.na(PIRADS)) %>%
  select(ID, Date_Biopsy, Prior_cancer, PSA, DRE, PIRADS, ISUP, Biopsy_type, Biopsy_method)%>%
  arrange(desc(Prior_cancer), PSA)

nrow(excl)

## [1] 54

# kable(excl, format = 'markdown')
# ID of potential exclusions
# print(excl$ID)

# Fiter data set ----
data <- data %>%
  filter(!(ID %in% excl$ID))

table(data$PIRADS, useNA = "always")

##
##      2      3      4      5 <NA>
##  59  275  676  323      0

data <- data %>%
  mutate(PIRADS = as.numeric(PIRADS),
         Systematic_cores_taken = as.numeric(Systematic_cores_taken, na.strings = "NA"),
         Target_cores_taken = as.numeric(Target_cores_taken, na.strings = "NA")
  ) %>% mutate( Cores = Systematic_cores_taken + Target_cores_taken)

# csPCa as ISUP >=2
data <- data %>%
  mutate(
    csPCa2 = ifelse(ISUP >= 2, TRUE, FALSE),
    log2_psa = log2(PSA),
    log2_vol = log2(Volume),
    log2_density = ifelse(!is.na(Volume), log2(PSA/Volume), NA),
    Age_cat = case_when(
      Age < 50 ~ "<50",
      Age < 60 ~ "50-59",
      Age < 70 ~ "60-69",
      Age < 80 ~ "70-79",
      Age >= 80 ~ "80-90",
    ),
    PSA_cat = case_when(
      PSA <= 3 ~ "[0,3]",
      PSA <= 4 ~ "(3,4]",
      PSA <= 5 ~ "(4,5]",

```

```

    PSA <= 6 ~ "(5,6]",
    PSA <= 7 ~ "(6,7]",
    PSA <= 10 ~ "(7,10]",
    PSA > 10 & PSA < 100 ~ "(10,100)",
    TRUE ~ NA)
)

# Define factor levels ----
data <- data %>% mutate(
  Prior_neg_biopsy = relevel(factor(Prior_neg_biopsy), ref="No")
  ,DRE = relevel(factor(DRE), ref="normal")
  ,Biopsy_type = factor(Biopsy_type, levels= c("Systematic", "Targeted only", "Targeted+Systematic"))
  ,Biopsy_method = relevel(factor(Biopsy_method), ref="Transperineal")
  ,ISUPcat = factor(ISUPcat, levels = c("ISUP<2" , "ISUP=2", "ISUP>2"))
  ,Age_cat = factor(Age_cat, levels = c("<50", "50-59", "60-69", "70-79", "80-90"))
  ,PSA_cat = factor(PSA_cat, levels = c("[0,3]", "(3,4]", "(4,5]", "(5,6]", "(6,7]", "(7,10]", "(10,100]"))
  ,PIRADScat2 = factor(PIRADScat2, levels = c("2-3", "4", "5"))
  ,PIRADScat3 = factor(PIRADScat3, levels = c("2-3", "4-5"))
  ,csPCa = factor(csPCa, levels = c("non-csPCa", "csPCa"))
)

```

Table 1. Clinical characteristics of the study sample and differences between non-csPCa and csPCa patients

```

data |>
  group_by(csPCa) |>
  tbl_summary(
    by = csPCa,
    include = c(Age,
      #Age_cat,
      #PSA_cat,
      PSA,
      Volume,
      PIRADS,
      #Prior_cancer,
      #Insurance,
      DRE,
      Prior_neg_biopsy,
      Biopsy_type,
      Biopsy_method,
      Cores
      #PIRADScat2,
      #ISUP,
      #Systematic_cores_taken,
      #Target_cores_taken
    ),
    missing_text = "Missing",
    label = list(
      Age_cat = "Age (years)",
      PSA_cat = "PSA",
      Prior_cancer = "Prior Cancer",
      Prior_neg_biopsy = "Prior negative biopsy",

```

```

    Biopsy_method = "Biopsy method",
    Biopsy_type = "Biopsy type",
    PIRADScat2 = "PIRADS category",
    Systematic_cores_taken = "Systematic cores",
    Target_cores_taken = "Target cores"
  ),
  type = all_continuous() ~ "continuous2", # Change to multi-line summary
  statistic = list(
    all_continuous() ~ c(
      "{median} ({p25}, {p75})", # First row
      "({min}, {max})"           # Second row (Min, Max)
    )
  )
)|>
# 1. Add the Overall (Total) column first (default behavior of last=FALSE)
add_overall() |>
# 2. Add the P-value column last
add_p()

# save as csv
# df_save <- data %>%
#   select(Age, PSA, Volume, DRE, PIRADS, Prior_neg_biopsy, ISUP) %>%
#   mutate(Prior_neg_biopsy = tolower(Prior_neg_biopsy)
#   ) %>%
#   rename(age = Age,
#           psa = PSA,
#           volume = Volume,
#           priornegbiopsy = Prior_neg_biopsy
#   )

# write.table(
#   x = df_save,
#   file = "C:/Users/ochernova/mydocs/pr2/MRIBiopsy21.01.2026.csv",
#   sep=",",
#   quote = FALSE,
#   row.names = FALSE
# )

```

Fit logistic regression for complete cases

```

data_filt <- data %>%
  filter(!is.na(DRE) & !is.na(Volume) & !is.na(PIRADS)
         & !is.na(Prior_neg_biopsy) & !is.na(Prior_neg_biopsy))

print(paste0("sample size complete cases ", nrow(data_filt)))

## [1] "sample size complete cases 1233"
print(paste0("csPCa ", nrow(data_filt[which(data_filt$csPCa == "csPCa"),])))

## [1] "csPCa 831"
print(paste0("percent csPCa ", nrow(data_filt[which(data_filt$csPCa == "csPCa"),])/ nrow(data_filt)))

```

```
## [1] "percent csPCa 0.673965936739659"
```

```
# df_comp <- data_filt %>%
#   select(Age, PSA, Volume, DRE, PIRADS, Prior_neg_biopsy) %>%
#   mutate(Prior_neg_biopsy = tolower(Prior_neg_biopsy),
#           id = row_number())
#   ) %>%
#   rename(
#     age = Age,
#     psa = PSA,
#     dre = DRE,
#     volume = Volume,
#     pirads = PIRADS,
#     priornegbiopsy = Prior_neg_biopsy
#   )
```

```
# write.table(
#   x = df_comp,
#   file = "C:/Users/ochernova/mydocs/webscr/datacomp.csv",
#   sep = ",",
#   row.names = FALSE,
#   quote = FALSE
# )
```

```
sumreg <- function(rg){
  t <- summary(rg)
  estim <- format(round(signif(t$coef[, 1], digits=2), 2), nsmall=2)
  HR <- format(round(exp(t$coef[, 1]), digits=2), 2, nsmall=2)
  HR.confint.lower <- format(round(exp(t$coef[, 1] + qnorm(0.025) * t$coef[,2]), digits=2), 2, nsmall=2)
  HR.confint.upper <- format(round(exp(t$coef[, 1] + qnorm(0.975) * t$coef[,2]), digits=2), 2, nsmall=2)

  HR <- paste0(HR, " (", HR.confint.lower, ", ", HR.confint.upper, ")")
  p.value<-signif(t$coef[, 4], digits=4)

  t2 <- data.frame(HR, p.value)#estim
  t2$p.value <- ifelse(t2$p.value < 0.001, '< 0.001',format(round(t2$p.value, 3), nsmall = 3))
  colnames(t2) <- c('OR (95% CI)', 'P value')
  return(kable(t2, format = 'markdown'))
}
```

```
# Fit the model
```

```
modell1 <- glm(formula = csPCa2 ~ Age + log2_psa + PIRADS + log2_vol + DRE + Prior_neg_biopsy,
               data = data_filt,
               family=binomial(link='logit'))
```

```
BIC(modell1)
```

```
## [1] 1114.112
```

```
round(summary(modell1)$coef[, 1],5)
```

##	(Intercept)	Age	log2_psa	PIRADS
##	-2.27918	0.05139	0.46219	1.31431
##	log2_vol	DREabnormal	Prior_neg_biopsyYes	

```
##           -1.19057           0.91551           -1.09070
summary(model1)

##
## Call:
## glm(formula = csPCa2 ~ Age + log2_psa + PIRADS + log2_vol + DRE +
##       Prior_neg_biopsy, family = binomial(link = "logit"), data = data_filt)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -2.27918    0.75929  -3.002  0.00268 **
## Age              0.05139    0.00943   5.450 5.04e-08 ***
## log2_psa        0.46220    0.10180   4.540 5.62e-06 ***
## PIRADS          1.31431    0.11938  11.010 < 2e-16 ***
## log2_vol       -1.19057    0.13375  -8.902 < 2e-16 ***
## DREabnormal     0.91551    0.21995   4.162 3.15e-05 ***
## Prior_neg_biopsyYes -1.09070  0.24126  -4.521 6.16e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1556.9  on 1232  degrees of freedom
## Residual deviance: 1064.3  on 1226  degrees of freedom
## AIC: 1078.3
##
## Number of Fisher Scoring iterations: 5
sumreg(model1)
```

	OR (95% CI)	P value
(Intercept)	0.10 (0.02, 0.45)	0.003
Age	1.05 (1.03, 1.07)	< 0.001
log2_psa	1.59 (1.30, 1.94)	< 0.001
PIRADS	3.72 (2.95, 4.70)	< 0.001
log2_vol	0.30 (0.23, 0.40)	< 0.001
DREabnormal	2.50 (1.62, 3.84)	< 0.001
Prior_neg_biopsyYes	0.34 (0.21, 0.54)	< 0.001

Imputation using mice method="pmm", m=5

```
datacov = data[,c("Age", "log2_psa", "log2_vol", "PIRADS", "DRE", "Prior_neg_biopsy", "csPCa2")]
na_counts <- colSums(is.na(datacov))
print(na_counts)

##           Age          log2_psa          log2_vol          PIRADS
##           0              0              81              0
##           DRE Prior_neg_biopsy          csPCa2
##           27              1              0

# Imputation
library(mice)
```



```

##
## Attaching package: 'mice'

## The following object is masked from 'package:stats':
##
##      filter

## The following objects are masked from 'package:base':
##
##      cbind, rbind

set.seed(2025)
impdata = mice(datacov, method="pmm", m=5)

##
## iter imp variable
## 1 1 log2_vol DRE Prior_neg_biopsy
## 1 2 log2_vol DRE Prior_neg_biopsy
## 1 3 log2_vol DRE Prior_neg_biopsy
## 1 4 log2_vol DRE Prior_neg_biopsy
## 1 5 log2_vol DRE Prior_neg_biopsy
## 2 1 log2_vol DRE Prior_neg_biopsy
## 2 2 log2_vol DRE Prior_neg_biopsy
## 2 3 log2_vol DRE Prior_neg_biopsy
## 2 4 log2_vol DRE Prior_neg_biopsy
## 2 5 log2_vol DRE Prior_neg_biopsy
## 3 1 log2_vol DRE Prior_neg_biopsy
## 3 2 log2_vol DRE Prior_neg_biopsy
## 3 3 log2_vol DRE Prior_neg_biopsy
## 3 4 log2_vol DRE Prior_neg_biopsy
## 3 5 log2_vol DRE Prior_neg_biopsy
## 4 1 log2_vol DRE Prior_neg_biopsy
## 4 2 log2_vol DRE Prior_neg_biopsy
## 4 3 log2_vol DRE Prior_neg_biopsy
## 4 4 log2_vol DRE Prior_neg_biopsy
## 4 5 log2_vol DRE Prior_neg_biopsy
## 5 1 log2_vol DRE Prior_neg_biopsy
## 5 2 log2_vol DRE Prior_neg_biopsy
## 5 3 log2_vol DRE Prior_neg_biopsy
## 5 4 log2_vol DRE Prior_neg_biopsy
## 5 5 log2_vol DRE Prior_neg_biopsy

# Model 1 on imputed data with PSA and Volume, PIRADS cont
fit2025m1 <- with(
  data = impdata,
  expr = glm(
    formula = csPCa2 ~ Age + log2_psa + PIRADS + log2_vol + DRE + Prior_neg_biopsy,
    family=binomial(link='logit')
  )
)

m1 <- summary(pool(fit2025m1))
round(m1[,2],5)

## [1] -2.36571 0.04983 0.47687 1.30155 -1.15838 0.92498 -1.09112

```

m1

##	term	estimate	std.error	statistic	df	p.value
## 1	(Intercept)	-2.36570739	0.739519410	-3.198979	1247.240	1.413857e-03
## 2	Age	0.04983099	0.008944129	5.571362	1318.244	3.060687e-08
## 3	log2_psa	0.47686845	0.098472921	4.842635	1278.113	1.437752e-06
## 4	PIRADS	1.30154944	0.116182493	11.202630	1266.649	7.672344e-28
## 5	log2_vol	-1.15837819	0.128876391	-8.988289	1029.564	1.176145e-18
## 6	DREabnormal	0.92497544	0.213452530	4.333401	1016.310	1.614024e-05
## 7	Prior_neg_biopsyYes	-1.09112326	0.232602691	-4.690931	1322.997	3.001486e-06

Characteristic	Overall N = 1,333 ¹	non-csPCa N = 436 ¹	csPCa N = 897 ¹	p-value ²
Age				<0.001
Median (Q1, Q3)	66 (58, 73)	61 (53, 69)	68 (60, 75)	
(Min, Max)	(33, 90)	(33, 88)	(42, 90)	
PSA				<0.001
Median (Q1, Q3)	7 (5, 10)	6 (5, 9)	7 (5, 11)	
(Min, Max)	(0, 99)	(0, 60)	(1, 99)	
Volume				<0.001
Median (Q1, Q3)	45 (35, 61)	50 (38, 75)	40 (32, 55)	
(Min, Max)	(6, 300)	(6, 300)	(6, 220)	
Missing	81	30	51	
PIRADS				<0.001
2	59 (4.4%)	49 (11%)	10 (1.1%)	
3	275 (21%)	186 (43%)	89 (9.9%)	
4	676 (51%)	178 (41%)	498 (56%)	
5	323 (24%)	23 (5.3%)	300 (33%)	
DRE				<0.001
normal	929 (71%)	394 (91%)	535 (61%)	
abnormal	377 (29%)	38 (8.8%)	339 (39%)	
Missing	27	4	23	
Prior negative biopsy	148 (11%)	82 (19%)	66 (7.4%)	<0.001
Missing	1	0	1	
Biopsy type				<0.001
Systematic	25 (1.9%)	16 (3.7%)	9 (1.0%)	
Targeted only	90 (6.8%)	13 (3.0%)	77 (8.6%)	
Targeted+Systematic	1,218 (91%)	407 (93%)	811 (90%)	
Biopsy method				0.088
Transperineal	452 (34%)	134 (31%)	318 (35%)	
Transrectal	881 (66%)	302 (69%)	579 (65%)	
Cores				0.5
Median (Q1, Q3)	15.0 (15.0, 16.0)	15.0 (15.0, 16.0)	15.0 (15.0, 16.0)	
(Min, Max)	(2.0, 22.0)	(2.0, 21.0)	(2.0, 22.0)	
Missing	20	7	13	

¹n (%)

²Wilcoxon rank sum test; Pearson's Chi-squared test